

📊 Customer Churn Prediction System – AI/ML Internship Task 04

🏢 Internship

****Organization:**** Devixo Solutions

****Task:**** AI/ML Internship Task 04 – End-to-End ML Application with Deployment

📌 Project Overview

This project is an end-to-end ****Customer Churn Prediction System**** developed as part of AI/ML Internship Task 04 at Devixo Solutions.

The system uses customer information to predict whether a customer is likely to ****Churn**** or ****Stay****. The complete machine learning workflow was implemented, starting from data preprocessing and feature engineering to model training, evaluation, hyperparameter tuning, model saving, and deployment through an interactive Streamlit application.

🎯 Objective

The main objective of this project was to build a complete machine learning application that:

- Processes a real-world customer dataset
- Handles missing values and duplicate records
- Detects and handles outliers
- Encodes categorical variables
- Performs feature engineering
- Scales numerical features
- Trains multiple machine learning models
- Compares model performance
- Applies cross-validation
- Performs hyperparameter tuning
- Saves the final trained model
- Provides an interactive prediction interface
- Deploys the application online

📁 Dataset

The project uses the ****Churn Modelling Dataset**** containing:

- **10,000 records**
- **14 original columns**

Important features include:

- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard
- IsActiveMember
- EstimatedSalary
- Exited (Target Variable)

🛠️ Data Processing

The following preprocessing steps were performed:

Dataset Check

- Dataset Shape: **10,000 × 14**
- Missing Values: **0**
- Duplicate Rows: **0**

Outlier Detection

Outliers were detected using the **Interquartile Range (IQR)** method.

Outliers detected:

Feature	Outliers
CreditScore	15
Age	359
Tenure	0
Balance	0
NumOfProducts	60
EstimatedSalary	0

Outliers were handled using **IQR Capping** instead of removing valuable customer records.

Categorical Encoding

Categorical variables were converted into numerical form using One-Hot Encoding.

Encoded features include:

- Geography_Germany
- Geography_Spain
- Gender_Male

After encoding, the dataset contained **12 features plus the target variable**.

Feature Engineering

A new feature called **BalanceSalaryRatio** was created.

This feature represents the relationship between a customer's account balance and estimated salary and can provide additional information for predicting customer churn.

Formula

`BalanceSalaryRatio = Balance / EstimatedSalary`

Feature Scaling

Numerical features were scaled using **StandardScaler** to ensure that features with different numerical ranges could be used effectively by machine learning algorithms.

Scaled Feature Shape:

10,000 × 12

Train/Test Split

The dataset was divided into training and testing sets using an **80/20 split**.

- Training samples: **8,000**
- Testing samples: **2,000**

🤖 Machine Learning Models

Four classification models were trained and compared:

1. Random Forest
2. Support Vector Machine (SVM)
3. Gradient Boosting
4. Decision Tree

📊 Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Initial Model Results

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	86.50%	79.65%	45.21%	57.68%	85.13%
SVM	86.15%	84.95%	38.82%	53.29%	82.31%
Gradient Boosting	87.25%	80.40%	49.39%	61.19%	86.80%
Decision Tree	79.75%	50.23%	53.56%	51.84%	70.00%

Based on the overall performance, **Gradient Boosting** performed best among the four models.

🔄 5-Fold Cross-Validation

5-Fold Cross-Validation was applied to validate model performance.

Model	Mean CV Accuracy	Standard Deviation
Random Forest	86.21%	0.0080
SVM	85.54%	0.0065
Gradient Boosting	**86.25%**	0.0086
Decision Tree	79.01%	0.0058

Gradient Boosting achieved the highest mean cross-validation accuracy.

🎯 Hyperparameter Tuning

****GridSearchCV**** was used to optimize the Gradient Boosting model.

Best Parameters

```
``text
learning_rate = 0.1
max_depth = 4
n_estimators = 100
```

Best Cross-Validation Accuracy

86.42%

The tuned Gradient Boosting model was selected as the final model.

🏆 Final Tuned Model Evaluation

The final tuned Gradient Boosting model achieved:

Metric	Score
Accuracy	86.90%
Precision	78.88%

Recall **48.65%**

F1 Score **60.18%**

ROC-AUC **87.01%**

Confusion Matrix

```
[[1540  53]
 [ 209 198]]
```

The final model provides a practical balance between overall accuracy and churn detection performance.

Model Saving

The final trained model was saved using **Joblib**.

models/best_model.joblib

This saved model is loaded by the Streamlit application to generate predictions without retraining the model.

Streamlit Application

An interactive web application was developed using **Streamlit**.

Users can enter:

- Credit Score
- Age
- Tenure
- Balance
- Number of Products
- Geography
- Gender
- Credit Card Status
- Active Member Status
- Estimated Salary

After clicking the prediction button, the application displays:

- Churn / Stay prediction
 - Churn probability
 - Customer input information
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Deployment

The Streamlit application was deployed online using **Streamlit Cloud**.

Live Demo

<https://devixosolution-aug6qd7gqwdzgzaevqt9fu.streamlit.app/>

GitHub Repository

https://github.com/shamii-101/DEVIXO_SOLUTION

Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - Seaborn
 - Joblib
 - Streamlit
 - Git
 - GitHub
 - Streamlit Cloud
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Project Structure

Task_04_AI_ML/

```
|
|— DataSet/
|   — Churn_Modelling.csv
|
```

```
├── models/
│   └── best_model.joblib
├── app.py
├── streamlit_app.py
├── requirements.txt
└── README.md
```

Key Learning Outcomes

Through this project, I gained practical experience in:

- End-to-end machine learning workflow
 - Data cleaning and preprocessing
 - IQR-based outlier detection and handling
 - Categorical encoding
 - Feature engineering
 - Feature scaling
 - Classification algorithms
 - Model comparison
 - Cross-validation
 - Hyperparameter tuning using GridSearchCV
 - Model evaluation
 - Model serialization using Joblib
 - Streamlit application development
 - Machine learning deployment using Streamlit Cloud
 - Git and GitHub project management
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Conclusion

The Customer Churn Prediction System successfully demonstrates a complete machine learning pipeline from raw data processing to online deployment.

Four machine learning models were trained and compared, followed by 5-Fold Cross-Validation and GridSearchCV hyperparameter tuning. Gradient Boosting was selected as the final model and achieved **86.90% test accuracy** and **87.01% ROC-AUC**.

The final trained model was integrated into a Streamlit application and deployed online, making the machine learning solution accessible through a web interface.



Project Status

Status: Completed 

AI/ML Internship Task 04 – Devixo Solutions